

Topics map in Basic Algebra:

classmate

Date _____

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1) Definition of Group & its properties

2) Subgroup

3) Group Homomorphism, its property

4) Group isomorphism, its property (Bijjective group Homomorphism)

5) Group automorphism (an isomorphism from G to G)

6) Kernel: ($\ker \phi$)

7) Image: ($\phi(G) \subseteq G'$ if $\phi: G \rightarrow G'$ be a group homo.)

8) Examples of Group: 1) Trivial group 2) $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ (under addition)

31) Transitive, faithful, free

32) Centraliser

33) Class Equation

34) Sylow's Theorem

35) Filtration of G 36) Simple group

3) Cyclic group $\leftarrow$ Finite Infinite 4) $(\mathbb{Z}/n\mathbb{Z})^*$ (multiplicative inverse)

5) $\text{Mor}(S, G)$

= collection of set maps $S \rightarrow G$

6) Product of groups: $G_1 \times G_2$

7) Boolean operation

8) $\text{Perm}(S)$, if S is not finite

if S is finite, $\text{Perm}(S) \cong S_n$

9) Universal Property

10) Boolean Operation

10) Cayley's Theorem

11) Examples of Homomorphism and Continuous homomorphism

12) Left and Right cosets

13) Lagrange's Theorem

14) Index of H in G

15) $|HK| = (|H||K|)/|H \cap K|$

16) Group of prime order is cyclic

17) Euler's Theorem

18) Group of prime order is cyclic, (finite)

19) Fermat's Little Theorem

20) Normal Subgroup & Example

9) Matrix Groups

10) Isometries of $\mathbb{R}^n$

11) Quaternion Group

12) Quaternions ($H = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$)

13) G, G' Abelian groups $\Rightarrow \text{Hom}(G, G')$

14) Free Groups & its Properties

15) Algebraic Topology

21) Conjugacy 22) Quotient Groups & canonical homo.

23) commutator subgroup 24) Abelianisation & free group

25) I, II, III isom Theorem

26) order of element 27) Torsion

28) Normalizer 29) Group actions & Types

30) Orbit, Stabiliser